

ArtiFlo: Mastering Business Process Modelling

Project Team

M. Shahmeer Durrani	21I-1164
Syed Nouman	21I-1172
Noman Mustafa Mehar	21I-1235

Session 2021-2025

Supervised by

Dr. Behjat Zuhaira



Department of Computer Science

**National University of Computer and Emerging Sciences
Islamabad, Pakistan**

June, 2025

Contents

1	Introduction	1
1.1	Background	1
1.2	BPMN and BPMN 2.0	2
1.2.1	Sample Process Description	2
1.2.2	Sample BPMN 2.0 XML Code	2
1.2.3	Sample BPMN 2.0 Diagram	4
1.3	Problem Statement	4
1.4	Motivation	5
1.5	Problem Solution	5
1.6	Stakeholders	6
1.6.1	Students	6
1.6.2	Educators/Instructors	7
1.6.3	Educational Institutions	7
1.6.4	Business Process Analysts	8
1.7	Product Users	8
1.7.1	Teachers (Educators/Instructors)	8
1.7.2	Students	9
2	Project Requirements	11
2.1	System Workflow Diagram	11
2.2	Use Cases	12
2.2.1	Use Case 1: AI-Powered BPMN Diagram Generation	12
2.2.2	Use Case 2: Manual BPMN Diagram Creation	13
2.2.3	Use Case 3: Real-Time BPMN Validation	14
2.2.4	Use Case 4: Real-Time Assistance and Feedback	15
2.2.5	Use Case 5: Activity Logging and Performance Monitoring	16
2.2.6	Use Case 6: Performance Reports	17
2.2.7	Use Case 7: Export/Import BPMN Diagrams	18
2.2.8	Use Case 8: AI Chatbot Support	19
2.3	Functional Requirements	20
2.3.1	Module 1: BPMN Diagram Creation	20
2.3.2	Module 2: Real-Time Assistance and Feedback	21

2.3.3	Module 3: Activity Logging and Performance Monitoring	22
2.4	Non-Functional Requirements	23
2.4.1	Reliability	23
2.4.2	Usability	24
2.4.3	Performance	25
2.4.4	Security	25
2.5	User Personifications (Personas)	26
2.5.1	Persona 1: The Undergraduate Student	26
2.5.2	Persona 2: The University Instructor	27
2.5.3	Persona 3: The Graduate Student	28
2.5.4	Persona 4: The Business Process Analyst	29
2.6	Use Case Diagram	29
2.7	Domain Model	30
3	System Overview	33
3.1	Architectural Design	34
3.1.1	Modular Program Structure	34
3.1.2	Cloud Architecture Overview	36
3.1.3	Future Evolution	36
3.2	Design Models	37
3.2.1	Activity Diagram	37
3.2.2	Class Diagram	38
3.2.2.1	Core Components Class Diagram	39
3.2.2.2	Supporting Services Class Diagram	39
3.2.3	Class-Level Sequence Diagrams	42
3.2.3.1	BPMN Diagram Creation	42
3.2.3.2	Course and Assignment Management	43
3.2.3.3	Feedback Management	44
3.2.3.4	Real-Time Feedback and Validation	44
3.2.3.5	Activity Logging and Reporting	45
3.2.3.6	User Authentication	45
3.2.3.7	Notifications Service	47
3.3	Data Design	47
3.3.1	Major Data Entities	47
3.3.2	ERD Visualization	49
3.3.3	Additional Considerations	49

List of Figures

1.1	Sample BPMN Diagram	4
2.1	Workflow Diagram for ArtiFlo	11
2.2	Use Case Diagram for ArtiFlo	30
2.3	Domain Model for ArtiFlo	31
3.1	Cloud Architecture	36
3.2	Activity Diagram for Business Process Modeling System	38
3.3	Core Components Class Diagram for ArtiFlo	41
3.4	Supporting Services Class Diagram for ArtiFlo	42
3.5	Class-Level Sequence Diagram for BPMN Diagram Creation	43
3.6	Class-Level Sequence Diagram for Course and Assignment Management .	43
3.7	Class-Level Sequence Diagram for Feedback Management	44
3.8	Class-Level Sequence Diagram for Real-Time Feedback and Validation .	44
3.9	Class-Level Sequence Diagram for Activity Logging and Reporting . . .	45
3.10	Class-Level Sequence Diagram for User Authentication	46
3.11	Class-Level Sequence Diagram for Notifications Service	47
3.12	Entity-Relationship Diagram (ERD) for ArtiFlo	50

List of Tables

- 1.1 Summary of Roles, Responsibilities, and Goals of Product Users 10
- 2.1 Use Case 1: AI-Powered BPMN Diagram Generation 12
- 2.2 Use Case 2: Manual BPMN Diagram Creation 13
- 2.3 Use Case 3: Real-Time BPMN Validation 14
- 2.4 Use Case 4: Real-Time Assistance and Feedback 15
- 2.5 Use Case 5: Activity Logging and Performance Monitoring 16
- 2.6 Use Case 6: Performance Reports 17
- 2.7 Use Case 7: Export/Import BPMN Diagrams 18
- 2.8 Use Case 8: AI Chatbot Support 19

Chapter 1

Introduction

The purpose of this project proposal is to present the development and deployment plan for the **ArtiFlo** system, an intelligent Business Process Modeling Notation (BPMN) tool that integrates Artificial Intelligence (AI) to support educational use. This system aims to provide a comprehensive learning environment for students and instructors, enhancing the teaching and learning of BPMN concepts through real-time feedback, automated diagram generation, and performance analysis.

1.1 Background

Business Process Model and Notation (BPMN) is a standard graphical representation used to model business processes. However, mastering BPMN can be challenging for students and educators due to the complexity of the notation and the intricacies involved in process modeling. Traditional BPMN tools often lack interactive educational features and real-time feedback, making it difficult for users to learn and apply BPMN concepts effectively.

To address these challenges, the **ArtiFlo** system leverages Large Language Models (LLMs) to automatically generate BPMN diagrams from natural language descriptions of business processes. Additionally, ArtiFlo provides real-time assistance during manual diagram creation by offering feedback on the correctness of the diagram's structure and flow. The system also logs user activities, enabling educators to monitor student progress and identify areas of difficulty. This data-driven approach aims to improve the overall learning experience, streamline BPMN education, and promote mastery of business process modeling.

1.2 BPMN and BPMN 2.0

Business Process Model and Notation (BPMN) is a standardized graphical notation used to visualize business processes. It provides an intuitive way for both business analysts and technical developers to model complex workflows and processes. BPMN allows organizations to create detailed process diagrams that are easy to understand and implement.

BPMN 2.0, an updated version of the original BPMN standard, introduces a comprehensive XML schema that defines the structure and semantics of BPMN models. This makes BPMN 2.0 not only a visual tool but also an executable one, enabling automated workflows and process management systems.

1.2.1 Sample Process Description

"If goods shall be shipped, the secretary clarifies who will do the shipping. If you have large amounts, special shipping will be necessary. In these cases the secretary invites three logistic companies to make offers and she selects one of them. In case of small amounts, normal post shipment is used. Therefore a package label is written by the secretary and a parcel insurance taken by the logistics department head if necessary. In the meantime the goods can be already packaged by the warehousemen. If everything is ready, the packaged goods are prepared for being picked up by the logistic company."

1.2.2 Sample BPMN 2.0 XML Code

Below is a sample of BPMN 2.0 XML code illustrating a simple process with a start event, a task, and an end event:

```
<bpmn:definitions xmlns:bpmn="http://www.omg.org/spec/BPMN
  /20100524/MODEL" xmlns:bpmndi="http://www.omg.org/spec/BPMN
  /20100524/DI" xmlns:di="http://www.omg.org/spec/DD
  /20100524/DI" xmlns:dc="http://www.omg.org/spec/DD
  /20100524/DC" xmlns:xsi="http://www.w3.org/2001/XMLSchema-
  instance" id="Definitions_1" targetNamespace="http://bpmn.
  io/schema/bpmn">
  <bpmn:collaboration id="Collaboration_0geryc1">
    <bpmn:participant id="Participant_0ygkg4f" name="Dispatch
      of goods; Computer Hardware Shop" processRef="
      Process_1" />
  </bpmn:collaboration>
  <bpmn:process id="Process_1" isExecutable="false">
    <bpmn:laneSet>
```

```

<bpmn:lane id="Lane_1viot5w" name="Logistics">
  <bpmn:flowNodeRef>Task_12j0pib</bpmn:flowNodeRef>
</bpmn:lane>
<bpmn:lane id="Lane_1ocseyo" name="Secretary">
  <bpmn:flowNodeRef>Task_0jsoxba</bpmn:flowNodeRef>
  <bpmn:flowNodeRef>Task_0vaxgaa</bpmn:flowNodeRef>
  <bpmn:flowNodeRef>Task_0e6hvnj</bpmn:flowNodeRef>
  <bpmn:flowNodeRef>Task_0s79ile</bpmn:flowNodeRef>
  <bpmn:flowNodeRef>InclusiveGateway_0p2e5vq</
    bpmn:flowNodeRef>
  <bpmn:flowNodeRef>InclusiveGateway_1dgb4sg</
    bpmn:flowNodeRef>
  <bpmn:flowNodeRef>StartEvent_1</bpmn:flowNodeRef>
  <bpmn:flowNodeRef>ParallelGateway_02fgrfq</
    bpmn:flowNodeRef>
  <bpmn:flowNodeRef>ExclusiveGateway_1mpgzhg</
    bpmn:flowNodeRef>
  <bpmn:flowNodeRef>ExclusiveGateway_1ouv9kf</
    bpmn:flowNodeRef>
</bpmn:lane>
<bpmn:lane id="Lane_1vl2igx" name="Warehouse">
  <bpmn:flowNodeRef>Task_05ftug5</bpmn:flowNodeRef>
  <bpmn:flowNodeRef>Task_0sl26uo</bpmn:flowNodeRef>
  <bpmn:flowNodeRef>ExclusiveGateway_0z5sib0</
    bpmn:flowNodeRef>
  <bpmn:flowNodeRef>EndEvent_1fx9yp3</bpmn:flowNodeRef>
</bpmn:lane>
</bpmn:laneSet>
<bpmn:sequenceFlow id="SequenceFlow_0iu9po7" name="no"
  sourceRef="ExclusiveGateway_1mpgzhg" targetRef="
    InclusiveGateway_0p2e5vq" />
<bpmn:inclusiveGateway id="InclusiveGateway_0p2e5vq">
  <bpmn:incoming>SequenceFlow_0iu9po7</bpmn:incoming>
  <bpmn:outgoing>SequenceFlow_1j94oja</bpmn:outgoing>
  <bpmn:outgoing>SequenceFlow_1dlbln9</bpmn:outgoing>
</bpmn:inclusiveGateway>
<bpmn:task id="Task_12j0pib" name="Insure_□parcel">
  <bpmn:incoming>SequenceFlow_1j94oja</bpmn:incoming>
  <bpmn:outgoing>SequenceFlow_0kz5g1t</bpmn:outgoing>
</bpmn:task>

```

This XML structure defines a simple process with a start event, a user task, and an end event, which can be executed or visualized using BPMN tools.

1.2.3 Sample BPMN 2.0 Diagram

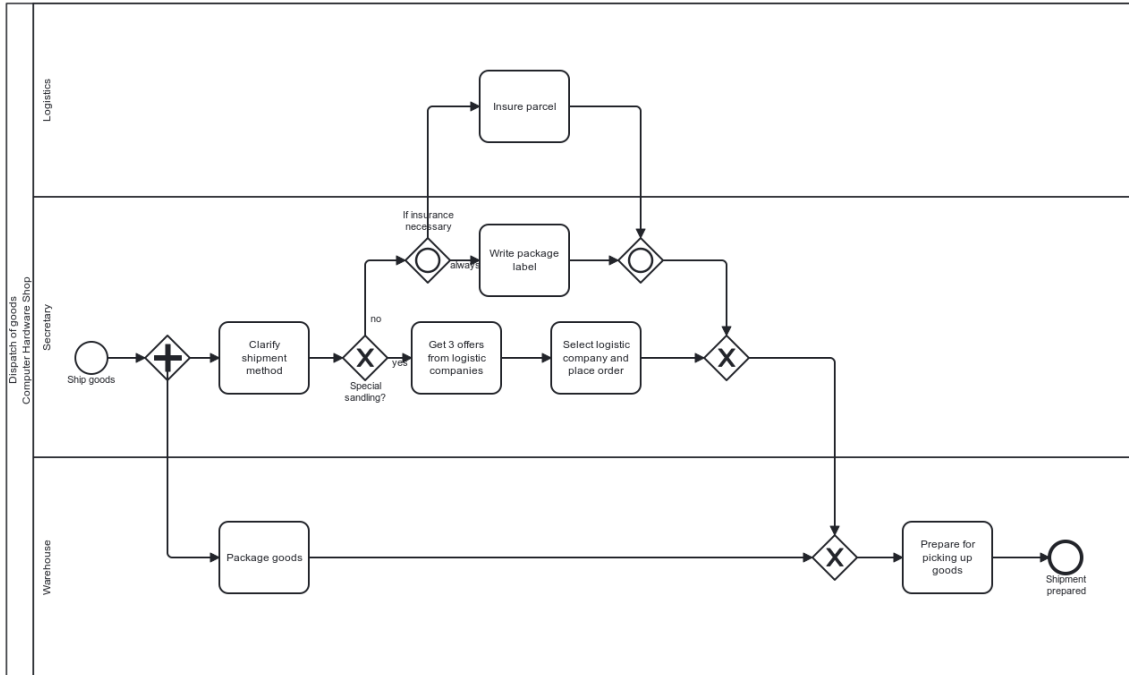


Figure 1.1: Sample BPMN Diagram

1.3 Problem Statement

The complexity of Business Process Model and Notation (BPMN) poses a significant challenge for students and educators alike, particularly in academic settings where understanding both the theory and application of BPMN is crucial. Traditional BPMN tools are typically designed for professional use and lack the educational features necessary to facilitate effective learning. They do not provide real-time feedback or guidance, making it difficult for students to identify and correct mistakes as they work through the modeling process. Educators, on the other hand, face difficulties in tracking student progress, identifying problem areas, and providing timely feedback.

ArtiFlo is being developed to address these challenges by integrating AI-driven features with BPMN diagram creation to enhance both the learning and teaching experience. The software automates BPMN diagram generation from natural language descriptions, provides real-time feedback to students during manual diagram creation, and offers comprehensive activity logging to track student performance. By doing so, ArtiFlo aims to transform inefficient BPMN learning environments into accessible, interactive, and data-driven systems that facilitate mastery of business process modeling.

1.4 Motivation

The motivation for this project stems from personal academic experiences and the observed challenges faced by students and educators in mastering Business Process Model and Notation (BPMN). As students of business process engineering, we encountered a lack of accessible resources and real-time feedback mechanisms essential for effective practical learning. The available BPMN tools are primarily geared toward professional users, lacking interactive guidance and educational features suited to academic environments.

Students often struggle to grasp BPMN concepts without immediate corrective feedback, leading to frustration and inefficiencies. Additionally, educators face challenges in creating multiple BPMN examples for instructional purposes and have limited means to identify specific areas where students may be struggling. These challenges hinder both the teaching and learning of BPMN, making it difficult to bridge the gap between theoretical knowledge and practical application.

By developing **ArtiFlo**, we aim to address these pain points by providing an AI-powered tool that simplifies BPMN diagram creation, delivers real-time feedback, and equips educators with insights into student performance. This project seeks to foster a more interactive, supportive, and efficient learning environment for mastering BPMN concepts.

1.5 Problem Solution

The software application being developed addresses the critical problem of inefficient and manual business process diagramming, which hampers the effectiveness of Business Process Management (BPM). The primary objective of ArtiFlo is to streamline and automate the creation of BPMN diagrams by leveraging advanced Natural Language Processing (NLP) and Large Language Models (LLMs). This approach will significantly reduce the time and effort required for diagramming, enhance accuracy, and facilitate better process documentation and analysis.

The key goals of the application include:

1. **Automated Diagram Generation:** Automatically generate BPMN diagrams from textual descriptions or process logs, minimizing manual input and errors.
2. **Intelligent Log Analysis:** Use LLMs to analyze logs of user activities within the BPMN diagrams, offering insights and recommendations for process improvements.

3. **Enhanced Accuracy:** Improve the precision of process diagrams by using advanced NLP techniques to interpret complex process descriptions.
4. **User-Friendly Interface:** Provide an intuitive interface that simplifies the creation, editing, and management of BPMN diagrams.
5. **Customizable Templates:** Offer customizable diagram templates catering to various business process needs, ensuring flexibility and adaptability.
6. **Real-Time Feedback:** Provide real-time feedback and suggestions for enhancing process diagrams based on user interactions and data patterns.
7. **Integration Capabilities:** Support integration with existing BPM tools and systems to facilitate seamless data exchange and process management.
8. **Error Detection:** Implement mechanisms to detect and correct inconsistencies or issues within the BPMN diagrams.
9. **Performance Tracking:** Track and analyze user performance and progress in creating BPMN diagrams, offering insights into productivity and areas for improvement.
10. **Scalability:** Ensure the application can scale to accommodate large and complex business processes, supporting the needs of various organizations.
11. **Security and Privacy:** Implement robust security measures to protect sensitive process data and ensure compliance with privacy regulations.
12. **Educational Support:** Provide educational resources and support to assist users in understanding and utilizing BPMN diagrams and the application's features.
13. **Continuous Improvement:** Incorporate user feedback and updates to continuously enhance the application's functionality and performance.

By achieving these goals, the application will significantly improve the efficiency and effectiveness of business process diagramming, leading to more streamlined and optimized processes.

1.6 Stakeholders

1.6.1 Students

Primary Users: Undergraduate and graduate students studying business process management, information systems, or related fields.

Needs:

- Easy-to-use tool for learning BPMN.
- Real-time feedback on diagram creation.
- Interactive tutorials.
- Performance tracking.

Goals:

- Master BPMN concepts.
- Create accurate process models.
- Improve diagramming skills.

1.6.2 Educators/Instructors

Key Users: University professors, lecturers, and instructors teaching business process modeling courses.

Needs:

- Efficient tool for creating BPMN diagrams.
- Ability to monitor student progress.
- Generate performance reports.
- Provide targeted feedback.

Goals:

- Streamline the BPMN teaching process.
- Improve student learning outcomes.
- Create multiple BPMN examples for course materials.

1.6.3 Educational Institutions

Decision Makers: Department heads, deans, or IT administrators responsible for adopting educational technologies.

Needs:

- Cost-effective, scalable solution that enhances business process modeling education.

Goals:

- Improve course offerings.
- Increase student engagement and performance in BPMN-related courses.

1.6.4 Business Process Analysts

Primary Users: Professionals analyzing and designing business processes to improve efficiency and effectiveness.

Needs:

- Advanced tool for creating, visualizing, and analyzing BPMN diagrams.
- Support for detailed process analysis and documentation.

Goals:

- Develop precise and insightful BPMN diagrams.
- Enhance process understanding.
- Provide valuable insights for business improvements.

1.7 Product Users

1.7.1 Teachers (Educators/Instructors)

Role: University professors and course instructors who use the tool to enhance the learning process for students.

Responsibilities:

- **Create and Manage BPMN Diagrams:** Use the tool to design and present BPMN diagrams for lectures, assignments, and projects.
- **Provide Feedback:** Offer guidance to students on their BPMN work.
- **Monitor Student Progress:** Track student performance through the tool's reporting features.

Goals:

- **Enhance Learning:** Use BPMN diagrams to clarify complex concepts.
- **Support Students:** Provide feedback to help students improve.
- **Facilitate Assignments:** Efficiently manage and grade BPMN assignments.

1.7.2 Students

Role: Students enrolled in business process management courses who interact with the tool for their coursework.

Responsibilities:

- **Create BPMN Diagrams:** Use the tool to create and submit BPMN diagrams as part of their coursework.
- **Participate in Assignments and Quizzes:** Complete assignments and take quizzes related to BPMN.
- **Seek Feedback:** Utilize teacher feedback to improve understanding and performance.

Goals:

- **Master BPMN:** Gain proficiency through hands-on practice.
- **Complete Assignments:** Successfully complete BPMN-related assignments and projects.
- **Receive Feedback:** Utilize feedback to enhance skills and knowledge.

User Type	Role	Responsibilities	Goals
Teachers	University professors and course instructors	- Create and manage BPMN diagrams - Provide feedback - Monitor student progress	- Enhance learning - Support students - Facilitate assignments
Students	Students enrolled in BPM courses	- Create BPMN diagrams - Participate in assignments and quizzes - Seek feedback	- Master BPMN - Complete assignments - Receive feedback

Table 1.1: Summary of Roles, Responsibilities, and Goals of Product Users

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the ArtiFlo system. The functional requirements detail the features and operations of the system, while the non-functional requirements cover the quality attributes that ensure the system performs effectively and meets users' expectations.

2.1 System Workflow Diagram

The System Workflow Diagram outlines the user flow and key interactions within the ArtiFlo platform. It demonstrates processes such as creating a BPMN diagram from scratch, receiving real-time feedback, and exporting the diagram. See Figure 2.1 for an overview of the workflow.

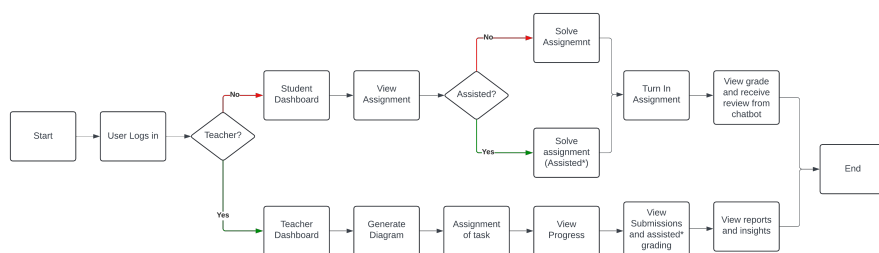


Figure 2.1: Workflow Diagram for ArtiFlo

2.2 Use Cases

2.2.1 Use Case 1: AI-Powered BPMN Diagram Generation

Attribute	Description
Actor	Instructor
Preconditions	- The instructor is logged into the system. - The system is ready to process natural language descriptions.
Basic Flow	1. The instructor selects the "AI-Powered BPMN Generation" feature from the dashboard. 2. Inputs a natural language description of a business process. 3. The system processes the input using its AI engine to generate a BPMN diagram. 4. The system displays the generated diagram for review. 5. The instructor can modify or confirm the generated diagram. 6. If modifications are needed, the instructor edits the diagram, and the system updates it accordingly.
Alternative Flow	If the system cannot accurately interpret the input: - The system requests more detailed input from the instructor. - The instructor revises the input, and the process continues from step 3.
Postconditions	- A BPMN diagram is generated and stored in the system. - The instructor can save, export, or further modify the diagram.
Special Requirements	- The AI engine must support multiple domains (e.g., healthcare, finance). - The system should handle ambiguous input by providing suggestions.

Table 2.1: Use Case 1: AI-Powered BPMN Diagram Generation

2.2.2 Use Case 2: Manual BPMN Diagram Creation

Attribute	Description
Actors	Student, Instructor
Preconditions	- The user (student or instructor) is logged into the system. - The drag-and-drop diagramming interface is available.
Basic Flow	1. The user selects "Manual BPMN Diagram Creation" from the dashboard. 2. The system opens the BPMN diagramming workspace with a blank canvas. 3. The user selects BPMN elements (tasks, events, gateways, etc.) from the library and places them on the canvas. 4. The user connects the elements using sequence flows. 5. The user arranges and organizes the diagram by dragging and snapping elements to the grid. 6. The user can undo/redo actions as necessary during the process.
Alternative Flow	If the user needs help with a specific BPMN element, they can consult the AI chatbot for real-time assistance.
Postconditions	- The manual BPMN diagram is saved in the system. - The user can export the diagram or continue working on it later.
Special Requirements	- The system must support undo/redo functionality. - The interface should provide tooltips explaining the use of each BPMN element.

Table 2.2: Use Case 2: Manual BPMN Diagram Creation

2.2.3 Use Case 3: Real-Time BPMN Validation

Attribute	Description
Actors	Student, Instructor
Preconditions	- The user is working on a BPMN diagram, either generated by the AI or created manually. - Real-time validation is enabled.
Basic Flow	1. As the user works on the BPMN diagram, the system continuously checks the diagram for compliance with BPMN 2.0 standards. 2. The system detects any errors (e.g., disconnected elements, missing end events, incorrect sequence flows). 3. The system highlights the error location in the diagram and displays a message explaining the issue. 4. The user clicks the highlighted error to view detailed suggestions for correction. 5. The user corrects the error based on the system's guidance, and the system validates the updated diagram.
Alternative Flow	The user can choose to disable real-time validation to focus on creating the diagram without interruptions.
Postconditions	- The BPMN diagram is validated and complies with BPMN 2.0 standards. - All errors are resolved, or the user has chosen to ignore validation warnings.
Special Requirements	- The system must provide detailed error messages with suggestions for correction. - Real-time validation should not slow down the diagramming process.

Table 2.3: Use Case 3: Real-Time BPMN Validation

2.2.4 Use Case 4: Real-Time Assistance and Feedback

Attribute	Description
Actors	Student, Instructor
Preconditions	- The user is logged in and working on a BPMN diagram. - The AI assistant is available to provide feedback.
Basic Flow	1. The user begins creating or modifying a BPMN diagram. 2. The system monitors the user's actions and provides feedback or suggestions in real-time. 3. The system analyzes the current diagram and predicts the next logical element (e.g., a gateway after a task). 4. The system offers a tooltip. 5. The user can accept or reject the suggestion.
Alternative Flow	If the user is unsure of a specific BPMN concept, they can consult the AI chatbot for guidance.
Postconditions	- The user receives continuous feedback on their diagram. - The diagram is enhanced by the system's suggestions.
Special Requirements	- The AI assistant must respond to diagram modifications in real-time without delays. - Predictive assistance should be context-sensitive and relevant to the user's current workflow.

Table 2.4: Use Case 4: Real-Time Assistance and Feedback

2.2.5 Use Case 5: Activity Logging and Performance Monitoring

Attribute	Description
Actor	Instructor
Preconditions	- The system is actively logging all user actions.
Basic Flow	1. The system logs every user action, including adding, editing, or deleting BPMN elements. 2. The system tracks the time spent on each diagram component. 3. The system aggregates the data and stores it for performance analysis. 4. The instructor can view detailed logs of student activities to monitor progress.
Postconditions	- All activities are logged and accessible by the instructor. - Logs are securely stored for future analysis.
Special Requirements	- The system must handle large volumes of log data without performance degradation. - Data should be securely stored and protected.

Table 2.5: Use Case 5: Activity Logging and Performance Monitoring

2.2.6 Use Case 6: Performance Reports

Attribute	Description
Actor	Instructor
Preconditions	- Activity logs and performance data are available for analysis.
Basic Flow	1. The instructor accesses the report generation feature from the teacher dashboard. 2. The system analyzes student activities, including time spent on tasks, revisions, and common mistakes. 3. The system generates a performance report that includes charts and graphs for easy interpretation. 4. The instructor reviews the report and identifies areas where students are struggling. 5. The instructor uses the report to offer personalized feedback to students.
Postconditions	- The instructor receives a detailed report of student performance. - Students receive targeted feedback based on their performance.
Special Requirements	- The reports must include visual aids (charts, graphs) to summarize data. - The system should allow the instructor to export reports in PDF or Excel formats.

Table 2.6: Use Case 6: Performance Reports

2.2.7 Use Case 7: Export/Import BPMN Diagrams

Attribute	Description
Actors	Student, Instructor
Preconditions	- The user has created or is working on a BPMN diagram.
Basic Flow	1. The user selects the export option from the diagramming interface. 2. The system offers a choice of formats (e.g., PNG, SVG, XML, BPMN 2.0 XML). 3. The user selects the desired format and downloads the diagram. 4. The user can also import an existing diagram by selecting the import option. 5. The system validates the imported diagram and allows the user to edit it.
Alternative Flow	If the imported diagram is incompatible, the system notifies the user and suggests corrective actions.
Postconditions	- The diagram is exported in the chosen format or successfully imported for editing.
Special Requirements	- The system must ensure compatibility between various BPMN formats. - Exported diagrams must retain their structure and visual quality.

Table 2.7: Use Case 7: Export/Import BPMN Diagrams

2.2.8 Use Case 8: AI Chatbot Support

Attribute	Description
Actor	Student
Preconditions	- The student is logged into the system and has access to the chatbot.
Basic Flow	1. The student initiates a conversation with the AI-powered chatbot by asking a question. 2. The chatbot processes the query and responds with relevant BPMN information or guidance. 3. The chatbot suggests next steps based on the student's current diagram state or query. 4. The student follows the guidance or asks follow-up questions as needed.
Postconditions	- The student receives helpful responses from the chatbot. - The chatbot assists the student in completing their BPMN diagram.
Special Requirements	- The chatbot must respond within 2 seconds of receiving a query.

Table 2.8: Use Case 8: AI Chatbot Support

2.3 Functional Requirements

The following are the core functional requirements of ArtiFlo, organized by major system modules. Each module represents a key feature area of the system.

2.3.1 Module 1: BPMN Diagram Creation

This module enables users to create and manipulate BPMN diagrams with ease.

1. **FR1: AI-powered BPMN generation:** The system shall enable instructors to generate BPMN diagrams by inputting natural language descriptions of business processes.
 - (a) **FR1.1:** The AI engine shall process the input and recognize standard patterns to generate a BPMN-compliant diagram.
 - **FR1.1.1:** The AI engine shall handle ambiguous process descriptions by offering suggestions for clarification.
 - **FR1.1.2:** The engine shall support business processes across various domains (e.g., finance, healthcare).
 - (b) **FR1.2:** The system shall provide users with an option to confirm or edit the auto-generated diagram.
 - **FR1.2.1:** Users shall be able to modify elements such as tasks, gateways, and events in the generated diagram.
 - **FR1.2.2:** The system shall provide recommendations for correcting errors or improving the process.
 - (c) **FR1.3:** The system shall allow for saving generated diagrams in progress to continue later.
2. **FR2: Manual BPMN creation interface:** The system shall provide a drag-and-drop interface that allows students to manually create BPMN diagrams, including all standard BPMN elements (e.g., tasks, events, gateways).
 - (a) **FR2.1:** The interface shall include a side panel with a library of BPMN elements (tasks, subprocesses, events, and gateways).
 - **FR2.1.1:** The library shall be categorized into types of elements (e.g., tasks, events, gateways) for easier access.
 - **FR2.1.2:** Hovering over elements in the library shall provide a tooltip explaining its use.

- (b) **FR2.2:** Users shall be able to place elements onto a canvas and connect them using sequence flows.
 - **FR2.2.1:** The system shall snap elements to a grid to maintain alignment.
 - **FR2.2.2:** Users shall be able to click and drag to rearrange the elements.
 - (c) **FR2.3:** The system shall allow users to undo/redo actions during diagram creation.
 - **FR2.3.1:** The undo/redo feature shall support up to 50 actions in sequence.
3. **FR3: Diagram modification tools:** Users shall be able to edit, resize, and delete BPMN elements in the diagram.
- (a) **FR3.1:** Users shall be able to click on an element to access a context menu with options to edit or delete the element.
 - (b) **FR3.2:** Users shall be able to resize certain elements, such as pools and lanes, by clicking and dragging their edges.
 - (c) **FR3.3:** Deleted elements shall trigger a confirmation prompt before removal.
4. **FR4: Real-time BPMN validation:** The system shall validate BPMN diagrams in real-time against BPMN 2.0 standards.
- (a) **FR4.1:** Validation errors shall be displayed to users with a description of the problem and suggestions for correction.
 - **FR4.1.1:** Errors shall include issues such as disconnected elements, missing end events, or incorrect sequence flows.
 - **FR4.1.2:** Users shall be guided on how to resolve the error by clicking on a tooltip associated with the error message.
 - (b) **FR4.2:** Users shall be able to disable real-time validation if desired to focus on diagram creation.

2.3.2 Module 2: Real-Time Assistance and Feedback

This module provides users with AI-driven feedback while creating BPMN diagrams to enhance learning.

1. **FR1: Real-time error detection:** The system shall detect structural errors in the BPMN diagram and notify users immediately, displaying visual cues and offering explanations.

- (a) **FR1.1:** The system shall highlight the error location with a red border around the problematic element.
 - (b) **FR1.2:** The system shall provide a pop-up message with a description of the error and suggestions for resolution.
- 2. **FR2: Predictive assistance:** The system shall offer predictive assistance, suggesting the next logical step or BPMN element based on the user's current diagram state.
 - (a) **FR2.1:** The system shall analyze the process flow and predict the most likely next element.
 - (b) **FR2.2:** Users shall have the option to accept or reject the suggestion, which will appear as a tooltip.
- 3. **FR3: AI chatbot support:** An AI-powered chatbot shall be available to provide contextual help related to BPMN elements, processes, and the system interface.
 - (a) **FR3.1:** The chatbot shall respond to user queries within 2 seconds.
 - (b) **FR3.2:** Users shall be able to ask the chatbot for specific help regarding BPMN notation and best practices.
 - (c) **FR3.3:** The chatbot shall integrate with external BPMN-related resources for additional guidance.

2.3.3 Module 3: Activity Logging and Performance Monitoring

This module tracks user activity for performance evaluation and feedback.

- 1. **FR1: Comprehensive activity logging:** The system shall log all user actions (e.g., creating, modifying, deleting elements) with timestamps.
 - (a) **FR1.1:** Activity logs shall include details of the action, the specific element, and the time of occurrence.
 - (b) **FR1.2:** Logs shall be stored securely in a database for future analysis.
- 2. **FR2: Time tracking:** The system shall track the time spent on each diagram component by individual students.
 - (a) **FR2.1:** Time spent per element shall be broken down and accessible by the instructor.
 - (b) **FR2.2:** Time tracking data shall be aggregated to provide an overall time spent per diagram.

3. **FR3: Performance metrics:** The system shall generate performance metrics based on logged activities.

(a) **FR3.1:** The system shall analyze user activity and generate reports that highlight areas of strength and areas for improvement.

(b) **FR3.2:** Reports shall include visual charts and graphs for easy interpretation.

2.4 Non-Functional Requirements

The non-functional requirements focus on the quality attributes that define the system's performance, usability, reliability, and security.

2.4.1 Reliability

- **NFR1: High system availability**

1. **NFR1.1:** The system shall maintain an uptime of 99.9 during peak usage times (e.g., class hours).

- **NFR1.1.1:** Uptime will be monitored continuously with real-time alerts for downtime.

- **NFR1.1.2:** Scheduled maintenance periods shall be communicated to users at least 24 hours in advance.

2. **NFR1.2:** System downtime (excluding planned maintenance) shall not exceed 10 minutes per month.

- **NFR1.2.1:** System recovery protocols shall ensure a recovery time objective (RTO) of less than 5 minutes.

- **NFR2: Data backup and recovery**

1. **NFR2.1:** The system shall perform automated backups daily to ensure no more than 24 hours of data is lost.

- **NFR2.1.1:** Backup verification shall be conducted weekly to ensure data integrity.

- **NFR2.1.2:** Backup data shall be stored in a secure, redundant cloud-based solution.

2. **NFR2.2:** In case of system failure, recovery procedures shall restore services within 15 minutes of detection.

- **NFR2.2.1:** The system shall prioritize recovery of critical services, including diagram creation and student progress tracking.

- **NFR3: Failure detection and alerting**

1. **NFR3.1:** The system shall detect critical failures and trigger alerts to system administrators within 2 minutes of an outage.
 - **NFR3.1.1:** Alerts shall be delivered through multiple channels, including email and SMS.
 - **NFR3.1.2:** An incident response protocol shall be followed to notify stakeholders and initiate troubleshooting.

2.4.2 Usability

- **NFR1: Intuitive user interface**

1. **NFR1.1:** The interface shall be designed to allow students and instructors to navigate core features (e.g., BPMN creation, feedback access) within 5 minutes of first use.
 - **NFR1.1.1:** User testing shall be conducted with students and instructors to ensure ease of use.
 - **NFR1.1.2:** Contextual tooltips shall guide users through complex actions (e.g., advanced BPMN elements).
2. **NFR1.2:** An onboarding tutorial shall be provided to new users, guiding them through the essential functions of the system.
 - **NFR1.2.1:** The tutorial shall be interactive and context-sensitive, adapting to user roles (student or instructor).

- **NFR2: Keyboard shortcuts**

1. **NFR2.1:** The system shall support keyboard shortcuts for common actions, such as undo, redo, and copy/paste.
 - **NFR2.1.1:** A reference list of shortcuts shall be accessible from the user interface.

- **NFR3: Accessibility features**

1. **NFR3.1:** The system shall comply with WCAG 2.1 standards for accessibility, ensuring compatibility with screen readers and high-contrast themes.
 - **NFR3.1.1:** All key features shall be navigable using only a keyboard.
 - **NFR3.1.2:** Visual elements shall have adequate contrast ratios, and alternate text descriptions for all media (e.g., icons, buttons).

2. **NFR3.2:** Users with disabilities shall be able to request custom accessibility features through their settings page.
 - **NFR3.2.1:** Customizations may include color adjustments, text size modifications, and speech-to-text input.

2.4.3 Performance

- **NFR1: Response time**

1. **NFR1.1:** The system shall provide real-time feedback with a response time of 1 second or less for user interactions.
 - **NFR1.1.1:** This includes feedback during BPMN diagram creation (e.g., error detection, predictive assistance).
 - **NFR1.1.2:** The system shall cache frequently accessed elements (e.g., BPMN shapes) to reduce load times.
2. **NFR1.2:** Latency shall not exceed 2 seconds during peak usage, even with up to 1000 concurrent users.
 - **NFR1.2.1:** Load balancing techniques shall be employed to distribute user requests evenly across servers.

- **NFR2: Scalability**

1. **NFR2.1:** The system shall scale horizontally to support up to 5000 concurrent users without degradation in performance.
 - **NFR2.1.1:** The system shall monitor and dynamically allocate resources based on usage patterns (e.g., peak usage hours).
2. **NFR2.2:** The system architecture shall support modular updates, allowing new features to be added without major downtime.

2.4.4 Security

- **NFR1: Data encryption**

1. **NFR1.1:** All data transmission shall be encrypted using TLS 1.2 or higher.
 - **NFR1.1.1:** The system shall enforce encryption for all user data, including login credentials, BPMN diagrams, and activity logs.
2. **NFR1.2:** Data at rest shall be encrypted using AES-256 encryption.
 - **NFR1.2.1:** This includes all user data stored in the database, including diagram data and performance metrics.

- **NFR2: User authentication**

1. **NFR2.1:** The system shall implement multi-factor authentication (MFA) for all users.
 - **NFR2.1.1:** MFA shall require a combination of a password and a time-based one-time password (TOTP) via a mobile app or SMS.
2. **NFR2.2:** Passwords shall be stored using a salted hash (e.g., bcrypt) to prevent reverse-engineering in case of a breach.
 - **NFR2.2.1:** The system shall prompt users to change passwords every 90 days.

- **NFR3: GDPR compliance**

1. **NFR3.1:** The system shall comply with GDPR regulations for user data protection and privacy.
 - **NFR3.1.1:** The system shall provide a consent mechanism for collecting user data.
 - **NFR3.1.2:** Users shall be able to request deletion of their personal data in compliance with the "right to be forgotten."
2. **NFR3.2:** Data processing agreements shall be in place with all third-party service providers.

2.5 User Personifications (Personas)

2.5.1 Persona 1: The Undergraduate Student

Name: Person A

Age: 21

Role: Undergraduate Business Management Student

Experience Level: Beginner in Business Process Modeling (BPMN)

Educational Background: Third-year business student who is just starting to learn about business process modeling in her course.

Goals:

- To master BPMN concepts through practice.
- To complete BPMN-related assignments and projects efficiently.
- To improve her BPMN diagramming skills with real-time feedback.

Frustrations:

- Finds it difficult to grasp all the BPMN elements.
- Often struggles with creating accurate BPMN diagrams.
- Feels overwhelmed with errors in her diagrams and would like more guidance.

Technology Comfort Level: Medium (comfortable with using online tools but not highly tech-savvy).

Usage Scenario: Person A logs in to ArtiFlo to complete a BPMN diagram for her course assignment (assigned to her by her instructor). She uses the manual BPMN creation feature to drag and drop elements. When she encounters errors, she relies on real-time feedback and the AI chatbot to guide her through corrections.

2.5.2 Persona 2: The University Instructor

Name: Person B

Age: 45

Role: Professor of Business Process Modeling

Experience Level: Expert in BPMN and Business Process Management

Educational Background: PhD in Information Systems and over 15 years of teaching BPMN.

Goals:

- To teach BPMN concepts efficiently to students.
- To monitor student progress and provide tailored feedback.
- To create multiple BPMN examples for course materials quickly.

Frustrations:

- Struggles to manage large classes and provide personalized feedback to every student.
- Finds it time-consuming to create BPMN diagrams manually.
- Wants a better way to track student progress and identify areas of difficulty.

Technology Comfort Level: High (well-versed in online tools and software platforms).

Usage Scenario: Person B uses ArtiFlo to generate BPMN diagrams for lectures. They leverage the AI-powered BPMN generation to quickly create diagrams from natural language descriptions. They then use the teacher dashboard to track the progress of their students and provide tailored feedback. For assignments, Person B views performance reports to identify which students are struggling with specific BPMN concepts.

2.5.3 Persona 3: The Graduate Student

Name: Person C

Age: 25

Role: Graduate Business Analyst Student

Experience Level: Intermediate in BPMN

Educational Background: Master's student in Business Process Management, has some experience with BPMN in her internships.

Goals:

- To enhance BPMN skills for real-world business scenarios.
- To get real-time feedback on complex BPMN diagrams.
- To learn more advanced BPMN techniques and optimize diagrams.

Frustrations:

- Sometimes struggles with complex process flows and deciding the right BPMN elements.
- Needs more advanced feedback on how to optimize diagrams beyond basic corrections.

Technology Comfort Level: High (very comfortable with technology and online tools).

Usage Scenario: Person C uses the real-time assistance and feedback feature of ArtiFlo to ensure her BPMN diagrams follow best practices. She often experiments with more advanced diagrams, and the system's assistance helps her understand the next steps for optimization. She uses the export/import functionality to share diagrams with classmates for collaborative projects.

2.5.4 Persona 4: The Business Process Analyst

Name: Person D

Age: 35

Role: Business Process Analyst at a large consulting firm

Experience Level: Expert in BPMN and process optimization

Professional Background: 10 years of experience in analyzing and optimizing business processes using BPMN and other methodologies.

Goals:

- To create precise BPMN diagrams for business clients.
- To streamline complex business processes by visualizing them in BPMN.
- To have a reliable tool that validates BPMN diagrams for real-time corrections.

Frustrations:

- Finds manual diagramming tedious for complex processes.
- Needs a tool that can validate BPMN diagrams and prevent errors during creation.
- Often works under tight deadlines, so speed and accuracy are essential.

Technology Comfort Level: High (advanced user of business process management tools).

Usage Scenario: Person D uses the AI-powered BPMN generation feature to quickly create diagrams from client descriptions of business processes. They rely on real-time validation to ensure the diagrams are error-free and comply with BPMN standards. Person D also appreciates the import/export functionality that allows sharing BPMN diagrams with clients or integrating them into other tools.

2.6 Use Case Diagram

The Use Case Diagram provides a visual representation of the interactions between users (students, instructors) and the functionalities of the ArtiFlo system. It illustrates how users can create and modify BPMN diagrams, access feedback, and generate reports. The actors represented in the diagram correspond to the user personas detailed in the previous section. Refer to Figure 2.2 for a detailed view.

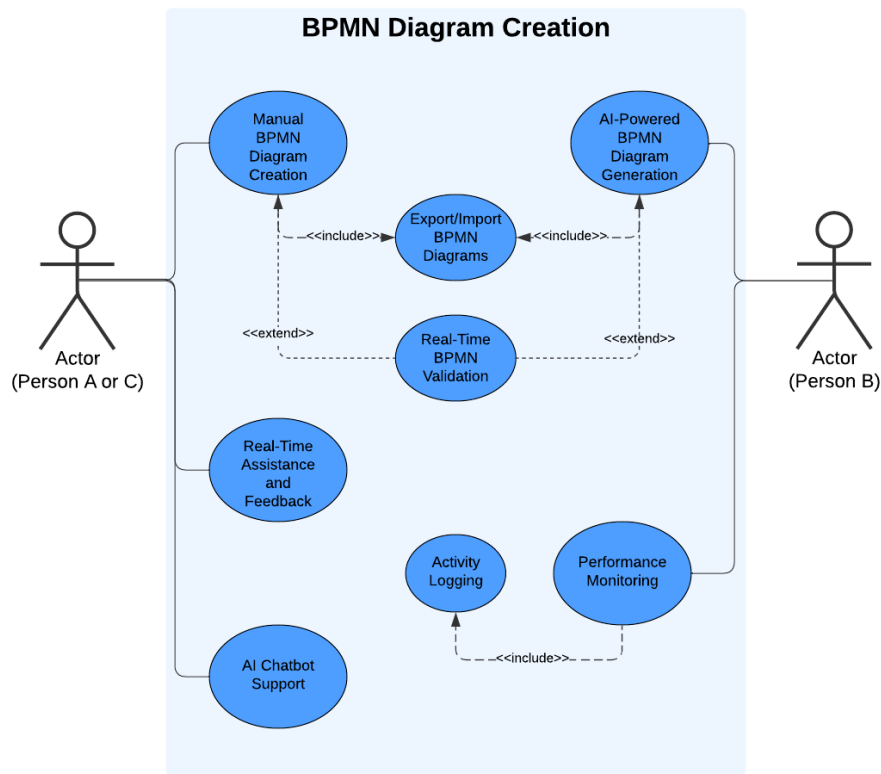


Figure 2.2: Use Case Diagram for ArtiFlo

2.7 Domain Model

The domain model for ArtiFlo illustrates the main entities and their relationships within the BPMN creation environment. The key entities are as follows:

- **User:** Represents system users, including both students and instructors.
 - **Student:** A user who creates BPMN diagrams and submits assignments for evaluation.
 - **Instructor:** A user who manages courses, assigns tasks, and provides feedback on student submissions.
- **Course:** A structured program of study that encompasses various BPMN assignments and is associated with instructors and students.
- **Assignment:** A specific task assigned by instructors to students, which requires the creation of BPMN diagrams.

- **Diagram:** A BPMN diagram created or edited by students, consisting of BPMN elements such as tasks, events, and gateways.
- **Activity Log:** Records each user's interaction with the system for tracking purposes, including actions taken on diagrams and assignments.
- **Performance Report:** A summary of performance metrics and feedback based on user activities, used by instructors to assess student progress and identify areas for improvement.
- **Feedback:** Input provided by instructors on submitted diagrams, highlighting strengths and areas needing revision.

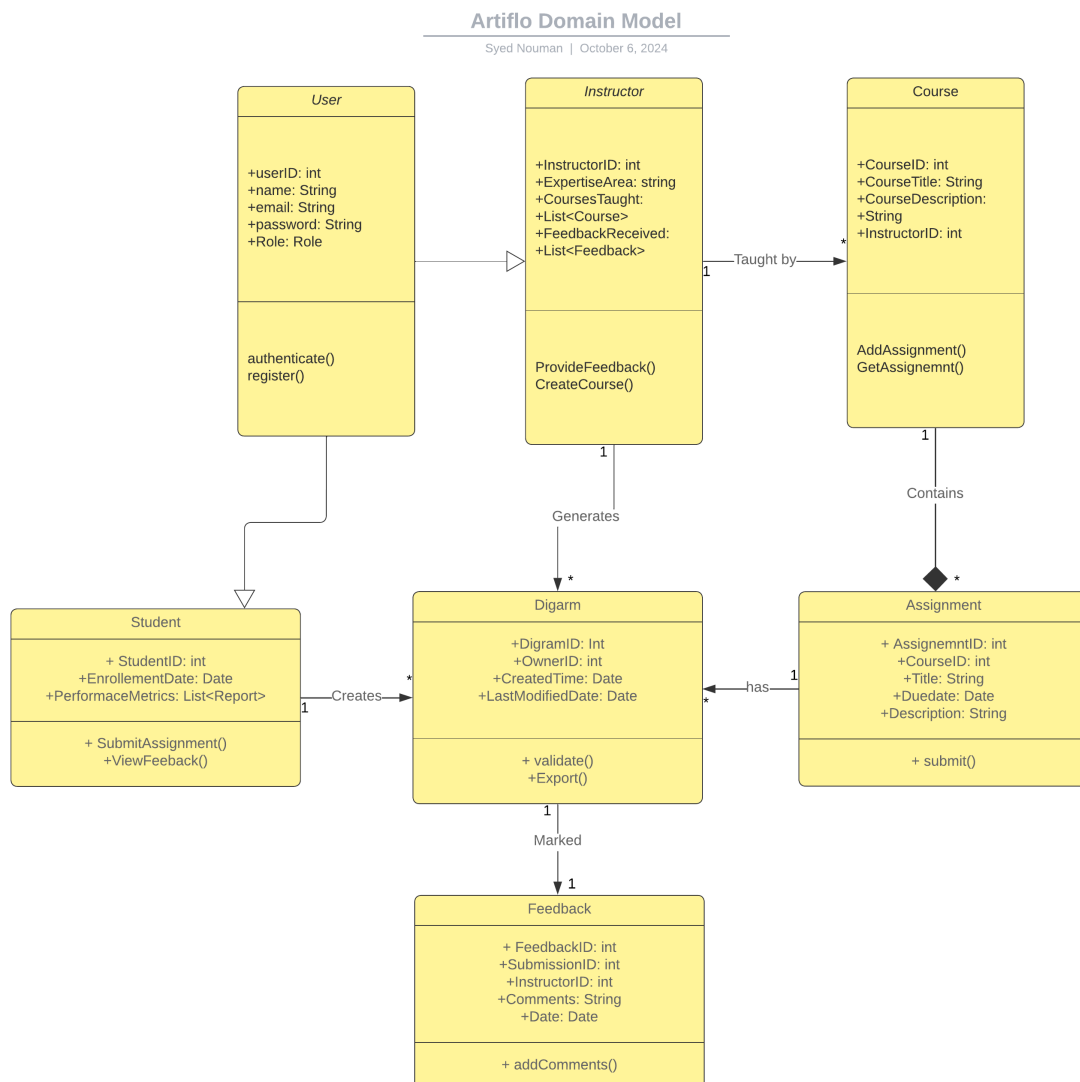


Figure 2.3: Domain Model for ArtiFlo

Chapter 3

System Overview

The "ArtiFlo: Mastering Business Process Modeling" system is a comprehensive web-based platform designed to enhance business process modeling education through the integration of advanced technologies like Large Language Models (LLMs), real-time validation, and diagram creation tools. It enables users (students and instructors) to create BPMN diagrams through natural language descriptions and manual drag-and-drop functionality, receiving instant feedback and validation on their work. The system is built with a modular architecture to ensure scalability, responsiveness, and ease of maintenance.

The system is composed of five major components:

- **LLM Microservice:** Responsible for AI-driven features, such as generating BPMN diagrams from natural language prompts. This service is deployed on a cloud server using Flask and Python libraries, providing the ability to interpret user inputs and deliver intelligent suggestions.
- **BPMN Linter and Validator:** A Node.js-based service that validates BPMN 2.0 XML files, ensuring compliance with BPMN standards and returning detailed reports on any detected errors or inconsistencies.
- **Main Server:** Built with ASP.NET Core and Blazor Pages, this server manages both server-side and client-side interactions. Blazor WebAssembly (WASM) enables high-performance client-side code execution, particularly for the diagramming interface, reducing latency and improving responsiveness.
- **Logs Analyzer:** This microservice is responsible for analyzing activity logs generated by users. It processes log data to identify patterns, detect anomalies, and provide insights into user behavior and system performance. This service aids in maintaining system reliability and improving user experience through data-driven decisions.

- **Performance Report Generator:** This microservice generates detailed performance reports for both students and instructors based on their activities within the system. It compiles metrics such as completion rates, assignment grades, and overall engagement, helping educators track student progress and improve instructional strategies.

The system also uses:

- **Syncfusion** for rich UI diagramming components.
- **Tailwind CSS** for modern, responsive frontend design.
- **SignalR** for real-time communication between the client and server.
- **PostgreSQL** and **Entity Framework Core** for data storage and management.

3.1 Architectural Design

3.1.1 Modular Program Structure

The architecture follows a **modular microservice-based design** to promote separation of concerns, scalability, and ease of deployment. Below is a high-level overview of the modules:

- **LLM Microservice:**
 - **Purpose:** Provides AI capabilities such as converting natural language to BPMN diagrams.
 - **Technology:** Hosted using Flask and Python on a cloud server.
 - **Communication:** Exposes an API that the main server calls when users request AI-generated BPMN diagrams or real-time feedback.
- **BPMN Linter and Validator:**
 - **Purpose:** Validates BPMN diagrams by analyzing BPMN 2.0 XML files and returning a list of any structural errors.
 - **Technology:** Built using Node.js.
 - **Communication:** Provides a RESTful API that the main server uses to validate diagrams submitted by users.
- **Main Server:**

- **Purpose:** The core hub that connects the frontend and backend components. Handles user requests, validates data, and orchestrates communication between other services.
- **Technology:** ASP.NET Core with Blazor Pages for both server-side rendering (SSR) and client-side rendering (CSR) via WASM.
- **Key Features:**
 - * **Blazor WebAssembly (WASM):** Allows client-side execution of C# code, improving responsiveness for complex tasks like diagram manipulation.
 - * **SignalR:** Provides real-time updates and communication, especially useful for AI feedback and validation tasks that require server interaction.
 - * **Syncfusion:** Powers the diagramming tool with extensive components for BPMN creation.
 - * **Tailwind CSS:** Ensures a responsive and modern UI.
 - * **Entity Framework Core:** Manages database operations with PostgreSQL.
 - * **AutoMapper:** Simplifies object mapping between different layers of the application.
- **Logs Analyzer:**
 - **Purpose:** Analyzes user activity logs to extract valuable insights and detect anomalies.
 - **Technology:** Built using a suitable data processing framework, potentially utilizing Python or Node.js for analytics.
 - **Communication:** Provides an API to deliver insights and analytics back to the main server or directly to users as needed.
- **Performance Report Generator:**
 - **Purpose:** Generates performance reports based on user activities, helping track student progress and instructor effectiveness.
 - **Technology:** Developed using a server-side language such as ASP.NET Core or Python.
 - **Communication:** Offers an API to facilitate report generation requests from the main server.
- **Database (PostgreSQL):**
 - **Purpose:** Stores user data, activity logs, diagrams, and performance reports.
 - **Technology:** Managed by PostgreSQL and accessed using Entity Framework Core.

3.1.2 Cloud Architecture Overview

The cloud-based infrastructure of the ArtiFlo platform is visualized in the diagram titled *ArtiFlo Cloud Architecture* (refer to Figure 3.1). This architecture is designed to ensure high availability, scalability, and security through the use of cloud-native technologies. Below is an overview of the main components in the diagram:

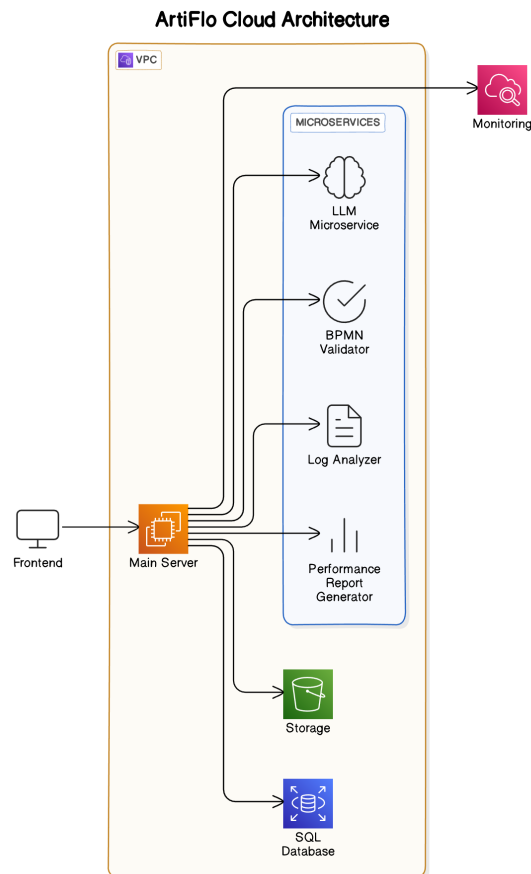


Figure 3.1: Cloud Architecture

3.1.3 Future Evolution

While this is the current architecture, the system is designed to be flexible. Additional microservices may be introduced in the future to support new features or enhance the performance of existing ones. Certain services may also evolve, be replaced, or be optimized as the needs of the system grow. The modular nature of the architecture ensures that such changes can be implemented with minimal disruption to the overall system.

3.2 Design Models

3.2.1 Activity Diagram

The following Activity Diagram illustrates the flow of interactions within the Business Process Modeling system. It represents the actions taken by instructors and students throughout the system, emphasizing the key processes involved in course management, assignment creation, BPMN diagram development, and feedback mechanisms.

- **Login:** Both the instructor and student log into the system. The system verifies their credentials.
- **Create Course:** The instructor creates a new course or selects an existing one for the students.
- **Assign Assignment:** The instructor creates and assigns a BPMN-related assignment to the students.
- **Select Course/Assignment:** The student selects the assigned course and BPMN assignment from their dashboard.
- **Create BPMN Diagram:** The student utilizes the BPMN diagramming tool to create their diagram.
- **Receive Real-Time Feedback:** The system provides real-time feedback to the student, identifying any syntax or semantic errors in their BPMN diagram.
- **Revise Diagram:** Based on the feedback received, the student revises their BPMN diagram.
- **Submit Diagram:** The student submits the finalized BPMN diagram for review by the instructor.
- **Monitor Student Progress:** The instructor views real-time progress reports or performance data of students working on the assignments.
- **Provide Feedback:** After reviewing the submitted diagrams, the instructor provides feedback to the students.
- **Log Activity:** The system logs every interaction made by both the instructor and student, including diagram creation, feedback received, and submission details.
- **Generate Reports:** The system generates performance reports for the instructor, summarizing student progress and areas for improvement.

- **Log Out:** Both the instructor and student log out of the system upon completion of their tasks.

Refer to Figure 3.2 for a visual representation of the system's activity flow.

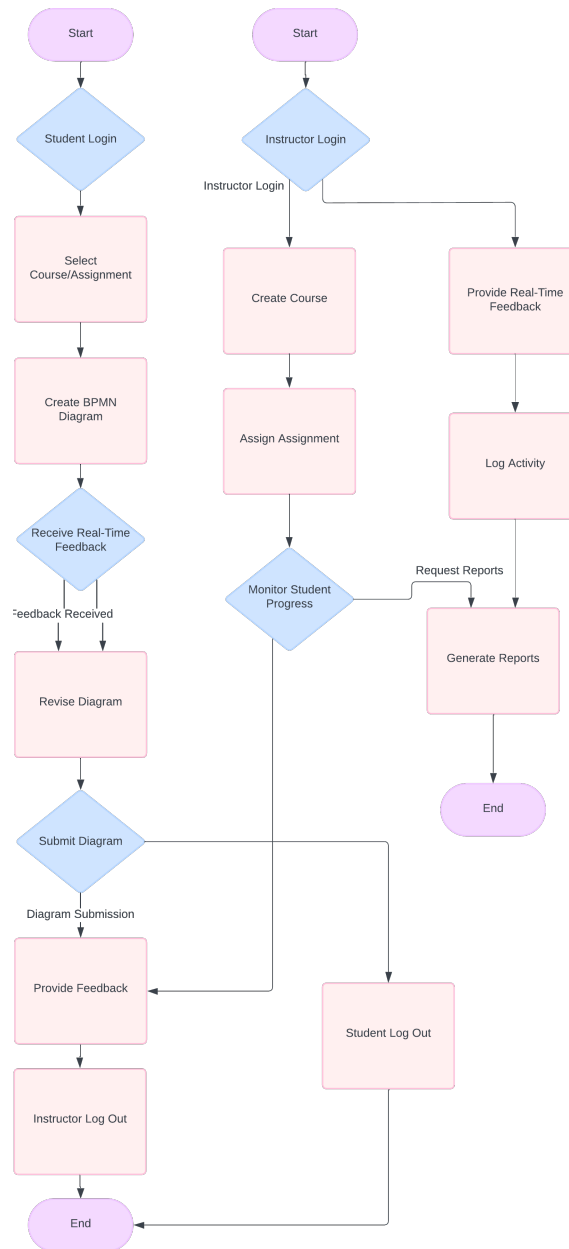


Figure 3.2: Activity Diagram for Business Process Modeling System

3.2.2 Class Diagram

The Class Diagram provides a visual representation of the system's structure, highlighting the relationships between various classes involved in the ArtiFlo project. For better readability and understanding, the Class Diagram has been divided into two distinct diagrams:

3.2.2.1 Core Components Class Diagram

The first class diagram focuses on the core components of the system, including entities related to the BPMN process and user interactions. This diagram encompasses the following classes:

- Course
- Assignment
- Student
- Instructor
- Feedback
- BPMNDiagram
- PerformanceReport
- ActivityLog
- Submission

This separation allows for a more straightforward view of the main functionalities and interactions within the ArtiFlo platform, emphasizing user roles and the core workflow related to business process modeling.

3.2.2.2 Supporting Services Class Diagram

The second class diagram encompasses the supporting services and infrastructure of the system. It includes:

- NotificationService
- ConfigurationService
- AuthenticationService
- UnitOfWork
- LoggingService
- Repositories:
 - UserRepository

- BPMNDiagramRepository
 - FeedbackRepository
 - SubmissionRepository
 - CourseRepository
 - AssignmentRepository
 - ActivityLogRepository
- Interfaces:
 - IUserRepository
 - IBPMNDiagramRepository
 - IFeedbackRepository
 - ISubmissionRepository
 - ICourseRepository
 - IAssignmentRepository
 - IActivityLogRepository

This diagram focuses on the technical architecture and operational services that support the core components, ensuring modularity and separation of concerns. By isolating these services, we enhance the maintainability and scalability of the system while allowing developers to focus on backend processes without cluttering the primary user-facing interactions.

Together, these two class diagrams provide a comprehensive view of the system's architecture, facilitating both high-level understanding and detailed technical insights. Refer to Figures 3.3 and 3.4 for detailed views of each diagram.

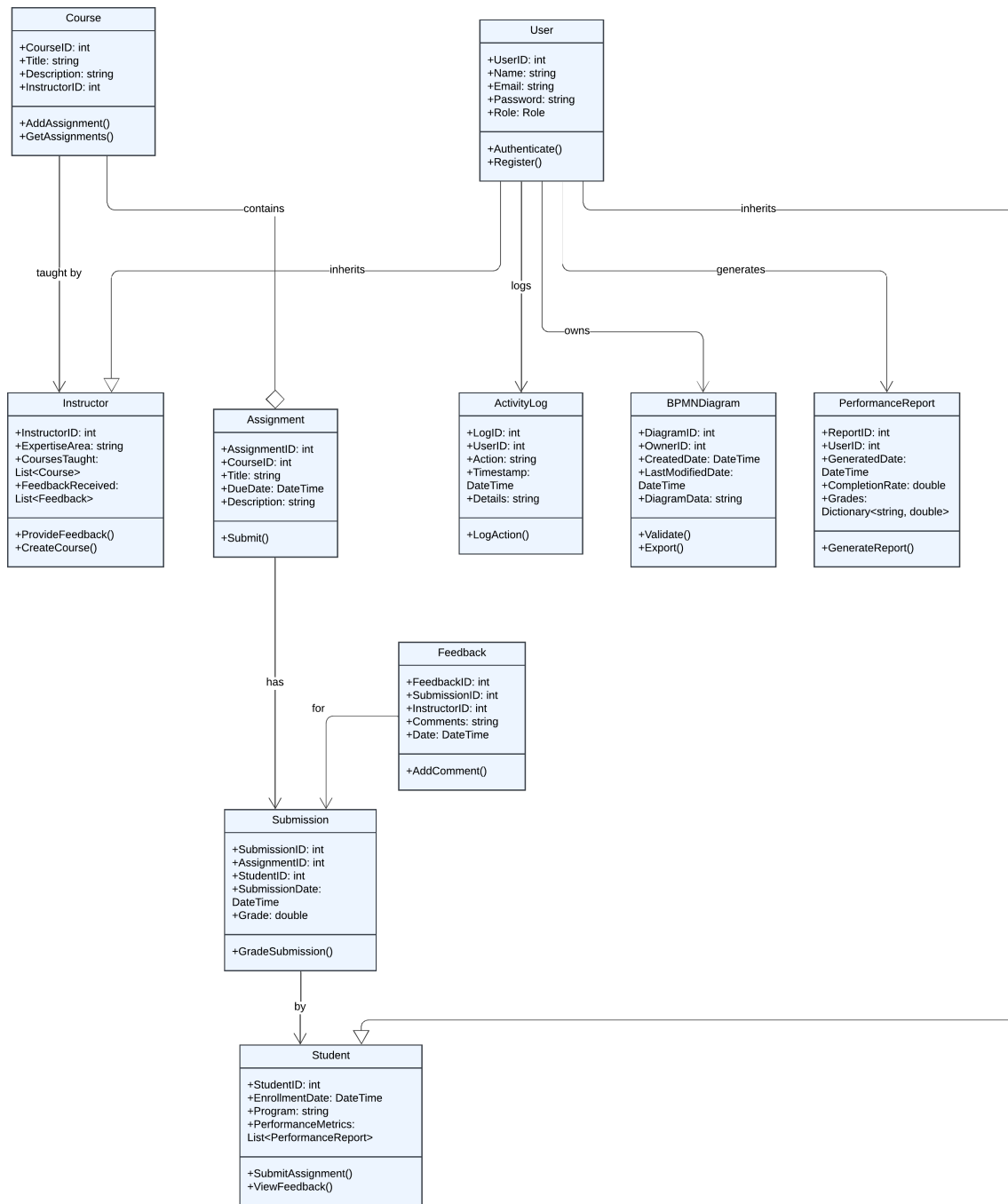


Figure 3.3: Core Components Class Diagram for ArtiFlo

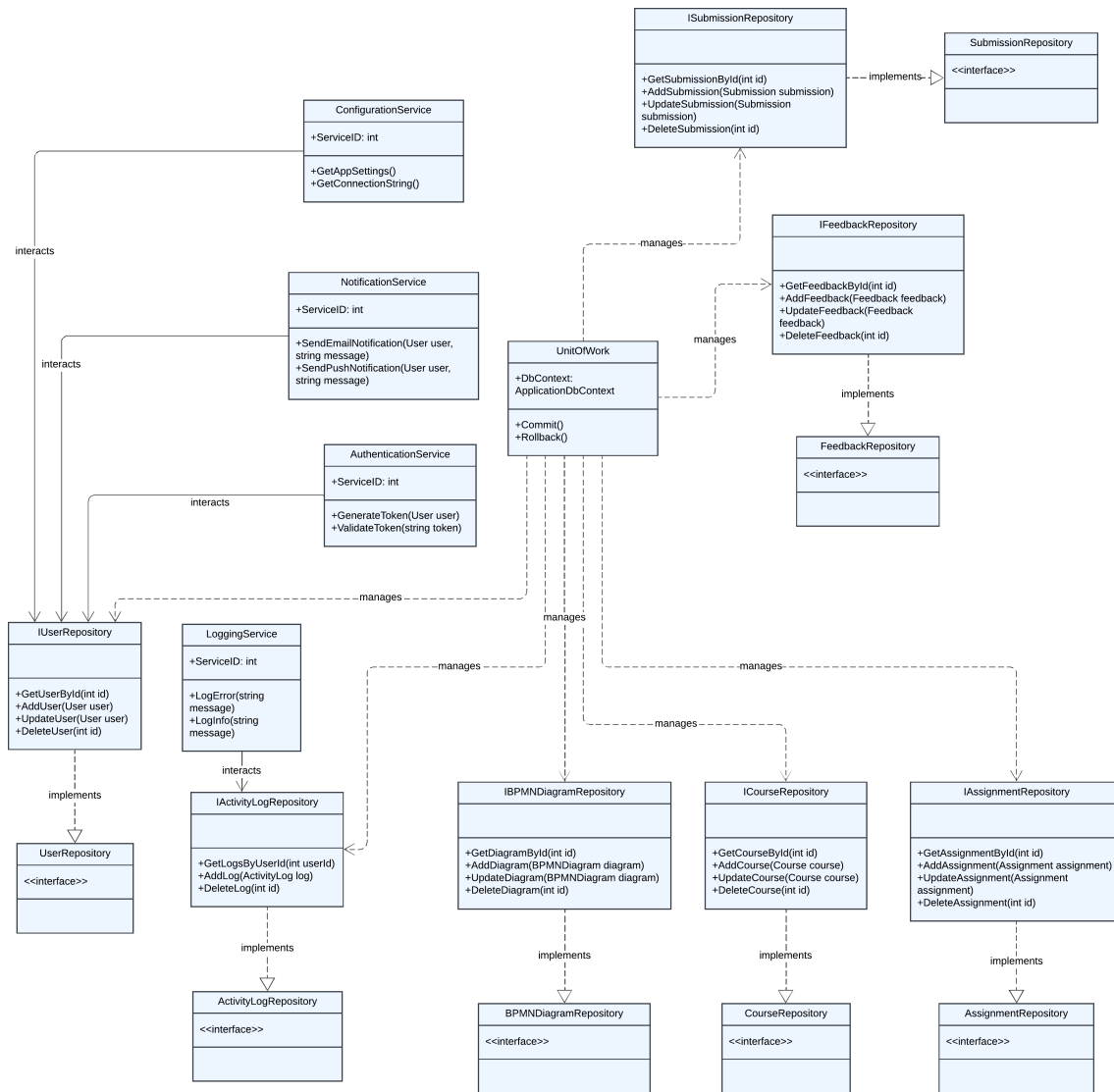


Figure 3.4: Supporting Services Class Diagram for ArtiFlo

3.2.3 Class-Level Sequence Diagrams

Class-level sequence diagrams provide a detailed view of the interactions between various objects and components of the ArtiFlo system. These diagrams help in understanding the flow of processes and the exchange of messages in response to specific events. The following class-level sequence diagrams capture key interactions within the system:

3.2.3.1 BPMN Diagram Creation

The BPMN Diagram Creation sequence illustrates how users interact with the system to generate BPMN diagrams, either through manual creation or AI-powered assistance. This

interaction involves the following components: the Student, Instructor, BPMNDiagram, BPMNRepository, and the BPMNValidator. Refer to Figure 3.5 for a detailed diagram.

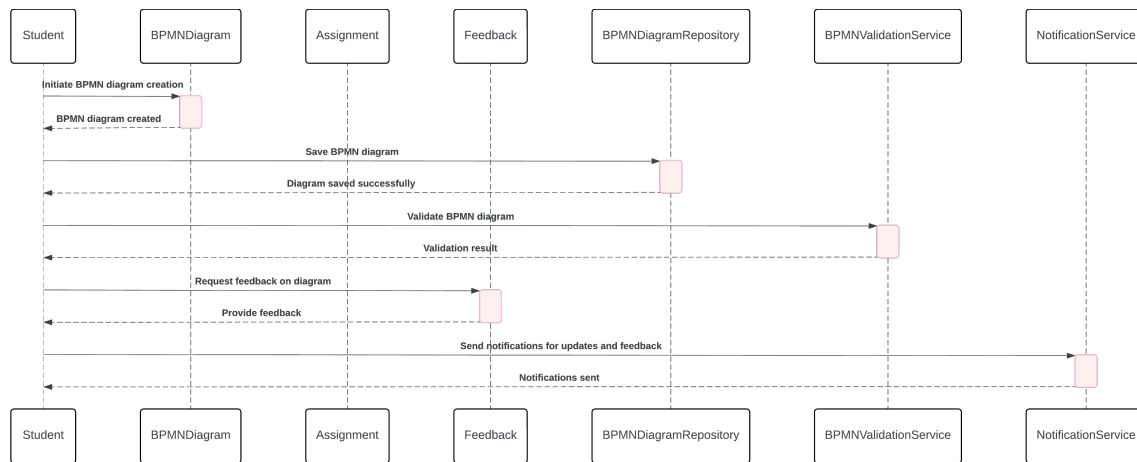


Figure 3.5: Class-Level Sequence Diagram for BPMN Diagram Creation

3.2.3.2 Course and Assignment Management

This diagram illustrates the flow of interactions between students, instructors, and the system for course and assignment management, including submission and grading. The primary components involved are Course, Assignment, Student, Instructor, and the relevant repositories. See Figure 3.6 for details.

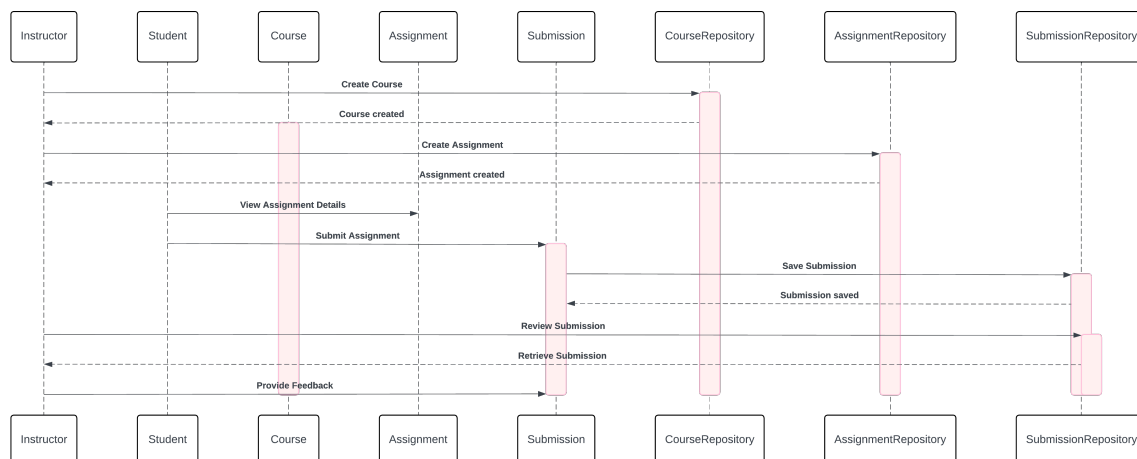


Figure 3.6: Class-Level Sequence Diagram for Course and Assignment Management

3.2.3.3 Feedback Management

This sequence diagram shows how feedback is managed within the system, from the instructor providing feedback to the student receiving it. The components involved are Instructor, Feedback, FeedbackRepository, and Student. Refer to Figure 3.7 for more details.

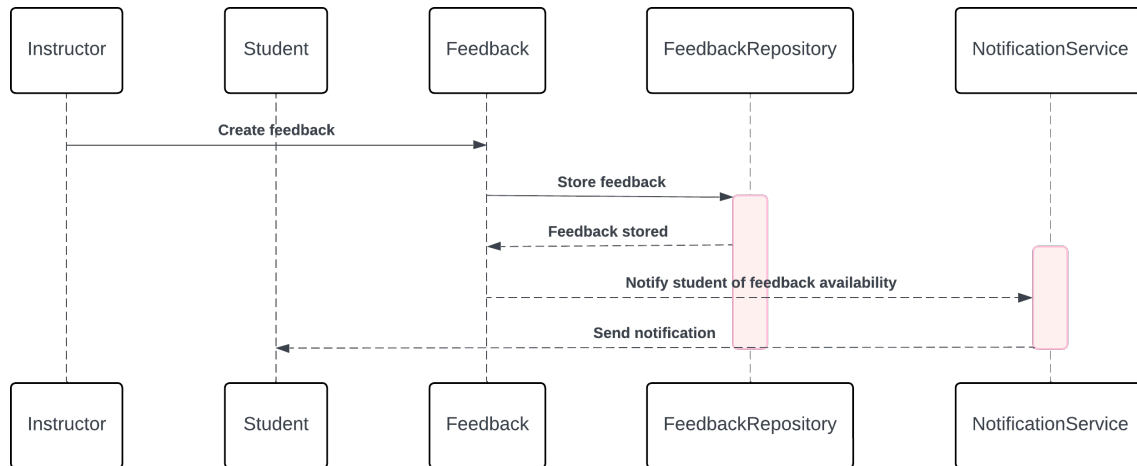


Figure 3.7: Class-Level Sequence Diagram for Feedback Management

3.2.3.4 Real-Time Feedback and Validation

The Real-Time Feedback and Validation diagram demonstrates how students receive instant feedback and validation while creating BPMN diagrams. This process includes interaction between Student, BPMNValidator, and the FeedbackRepository. Figure 3.8 provides the detailed interaction flow.

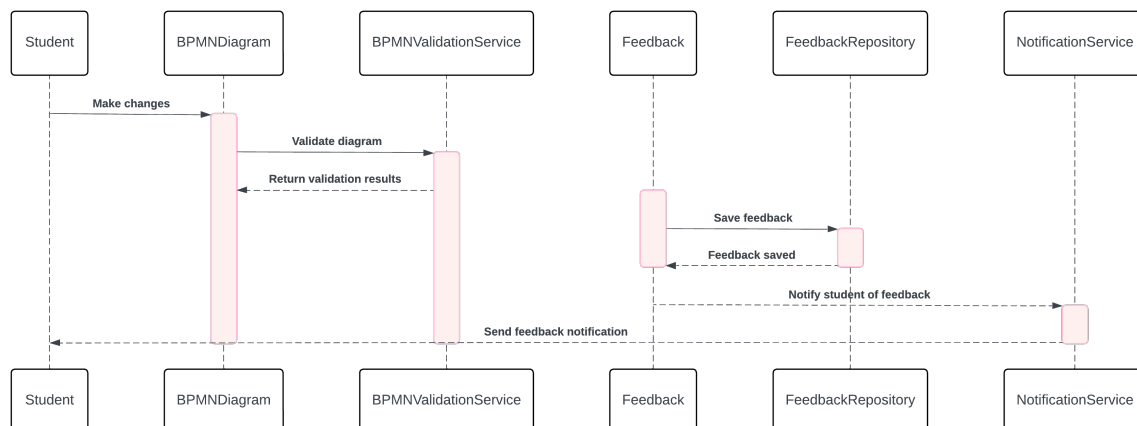


Figure 3.8: Class-Level Sequence Diagram for Real-Time Feedback and Validation

3.2.3.5 Activity Logging and Reporting

This diagram outlines how activity logs are generated and used to create performance reports for users. The interaction includes ActivityLog, PerformanceReport, Instructor, and Student. Refer to Figure 3.9 for the diagram.

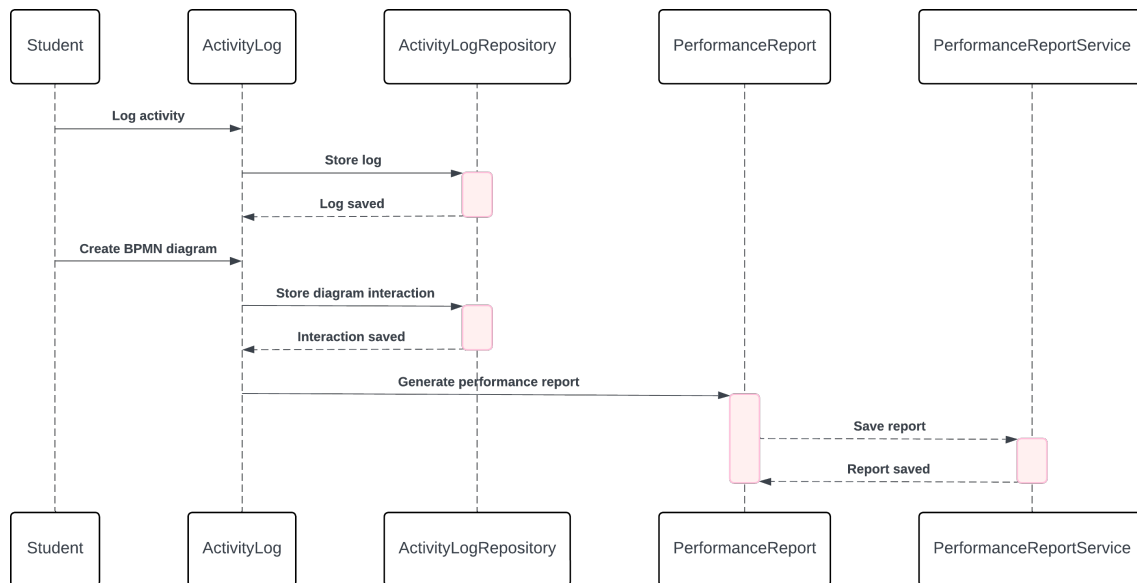


Figure 3.9: Class-Level Sequence Diagram for Activity Logging and Reporting

3.2.3.6 User Authentication

The User Authentication sequence diagram demonstrates the process of authenticating users within the system. This interaction includes User, AuthenticationService, and UserRepository. See Figure 3.10 for a detailed diagram.

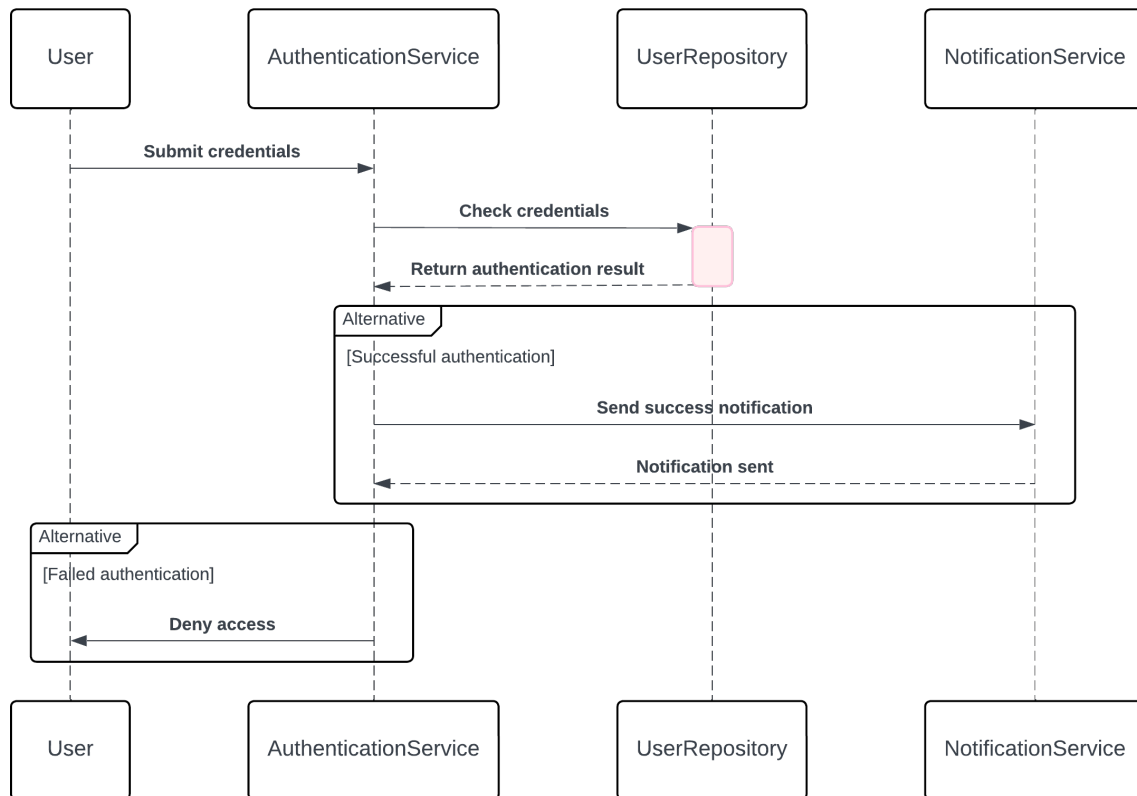


Figure 3.10: Class-Level Sequence Diagram for User Authentication

3.2.3.7 Notifications Service

The Notifications Service sequence diagram details how notifications are generated and sent to users. This includes interactions between User, NotificationService, and NotificationRepository. Figure 3.11 illustrates the process.

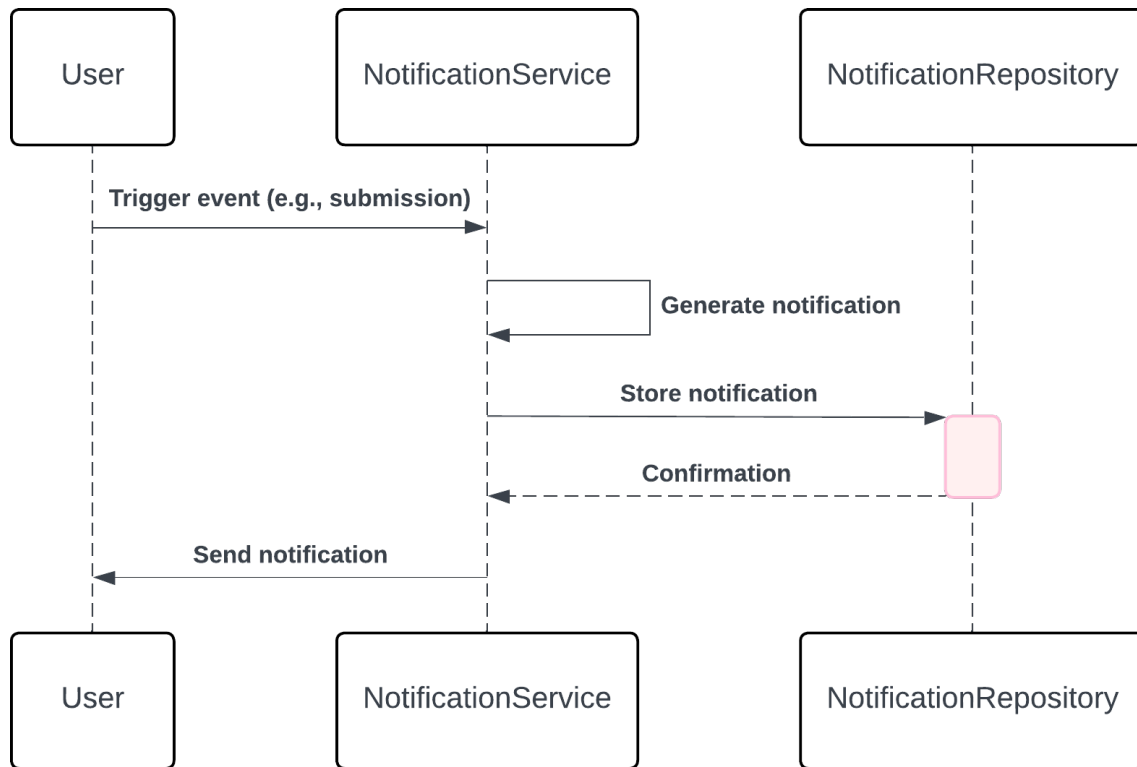


Figure 3.11: Class-Level Sequence Diagram for Notifications Service

3.3 Data Design

The data design of the "ArtiFlo: Mastering Business Process Modeling" system is crucial for effectively managing and processing the information generated by users within the platform. The system utilizes a relational database model, specifically PostgreSQL, to store and organize its data. The following sections outline the key entities in the system and how they relate to each other.

3.3.1 Major Data Entities

- User

- The User table is central to the system, storing information about all users, including students and instructors. Key attributes include UserID, Name, Email, Password, and Role, allowing for authentication and role-based access control.
- Email verification and authentication mechanisms are not detailed in the ERD but are vital for user registration and security. Typically, a separate AuthKeys table would store tokens for email verification, password resets, and other security-related processes.

- **Student**

- The Student table is linked to the User table through the StudentID, which serves as a foreign key. This table captures specific attributes relevant to students, such as Major, YearOfStudy, and GPA, enabling personalized learning experiences and tracking academic performance.

- **Instructor**

- Similar to the Student entity, the Instructor table is associated with the User table. It stores information specific to instructors, including their department and office hours, facilitating communication and resource allocation for students.

- **Course**

- The Course table maintains details about various courses offered within the platform. Each course is linked to an instructor through the InstructorID, ensuring that each course can be properly assigned and managed.

- **Assignment**

- The Assignment table records the assignments related to each course. It contains information such as DueDate and Description, facilitating assignment management and tracking for both students and instructors.

- **Submission**

- The Submission table captures students' submissions for assignments. Each submission is linked to a specific student and assignment, along with attributes like SubmissionDate and Grade, allowing instructors to evaluate student performance effectively.

- **BPMN Diagram**

- This table stores the BPMN diagrams created by users. Each diagram includes a title, description, and XML data representing the diagram, ensuring that users can manage and retrieve their diagrams efficiently.
- **Feedback**
 - The Feedback table links instructor feedback to specific submissions, allowing for tailored assessments and comments that can guide student learning.
- **Performance Report**
 - This table is essential for tracking student performance in courses, storing metrics related to their progress and achievements.
- **Activity Log**
 - The Activity Log records user actions within the platform, facilitating auditing and monitoring of user interactions.

3.3.2 ERD Visualization

The following figure illustrates the relationships among the various entities within the system, showing how data is interconnected and structured:

3.3.3 Additional Considerations

While the current ERD includes the primary entities relevant for understanding the system's operation, it does not encompass all necessary components for a fully functioning web application. For instance, aspects related to email verification, authentication keys, and security measures are critical but not depicted here. In a complete system design, additional tables such as `AuthKeys` or `Notifications` for managing verification tokens, password recovery mechanisms, and user session management would be integrated to enhance security and user experience.

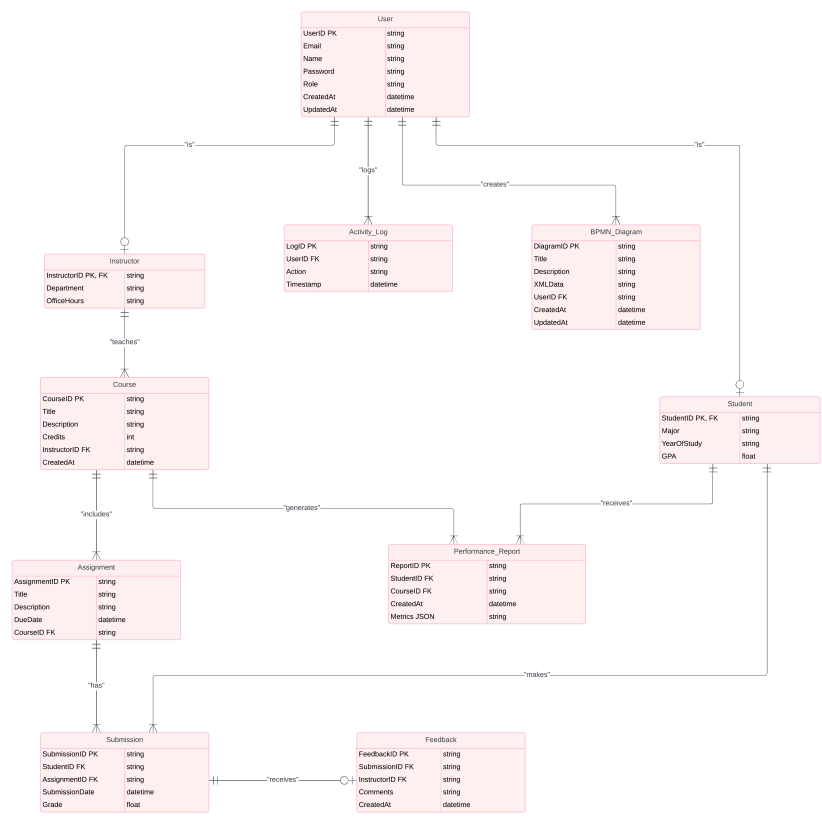


Figure 3.12: Entity-Relationship Diagram (ERD) for ArtiFlo